

ROAD HEALTH & CITIZEN GRIEVANCE REPORT

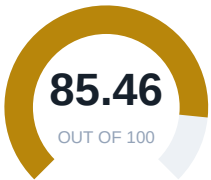
Automated computer-vision assessment of road surface condition, defect extent and remediation urgency, prepared for municipal grievance submission.

CONDITION: FAIR

RISK: MODERATE · Priority-3

REF: RHA-20260903-A7F3C1

ROAD HEALTH INDEX



FAIR

DEFECTS DETECTED

5

3 potholes, 2 alligator cracks

DEFECT AREA RATIO

14.54%

of the photographed surface

HIGHEST SEVERITY

High

Pothole - tag #1

ROAD RISK ASSESSMENT

Project-defined model - see methodology

ROAD RISK INDEX

32.4

MODERATE RISK

Priority-3

Remediate within 30 days

REMEDIATION URGENCY

32.4 / 100



Extent		58 /100	14.54% of the frame damaged weight 50%
Worst defect		61 /100	6.08% of the frame in one box weight 30%
Density		62 /100	5 defects detected weight 20%

Weighted mean of the components above, scaled by a hazard factor of 0.89 (how dangerous the detected classes are, area-weighted) and a confidence factor of 0.61 (mean confidence, floored at 0.50).

GEOSPATIAL & CAPTURE TELEMETRY

Date & time	03 Sep 2026, 03:16 PM
Coordinates	17.494800° N, 78.399600° E
Region	Hyderabad, Telangana, India
Source image	road_photo.jpg (1600 x 1000 px)
Evidence SHA-256	af2bdb1aa9b6ec1e2ade1d6...
Map link	Open this location in Google Maps

MUNICIPAL JURISDICTION & ROUTING

Recipient body	Greater Hyderabad Municipal Corporation (GHMC)
Grievance tier	Priority-3 - Remediate within 30 days
Dominant defect	Pothole
Priority action	Hot-mix / cold-mix asphalt fill with mechanical compaction
Submission status	Prepared by the citizen - not yet filed with the authority

Official Citizen Grievance Statement

Road surface defects were detected in a photograph captured at latitude 17.494800, longitude 78.399600. Automated analysis identified 3 potholes and 2 alligator cracks, covering approximately 14.54% of the photographed road surface. The computed Road Health Score is 85.46 out of 100, indicating a FAIR road condition. The estimated Road Risk Index is 32.4 out of 100, placing this location in the MODERATE risk band; the suggested grievance tier is Priority-3 (remediate within 30 days). The most serious individual defect is rated high severity. The Greater Hyderabad Municipal Corporation (GHMC) is kindly requested to inspect the location and undertake the necessary repair work.

Addressed to: Greater Hyderabad Municipal Corporation (GHMC)

DEFECT INVENTORY & RECOMMENDED REMEDIATION

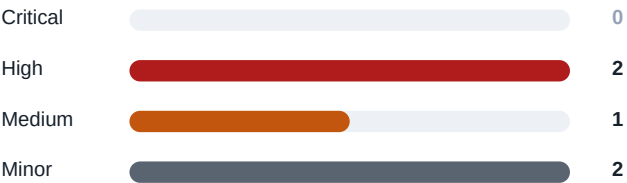
Suggestions for the engineering wing, not directives

TAG	CLASSIFICATION	DETECTION CONFIDENCE		SEVERITY	FRAME AREA	RECOMMENDED REMEDIATION
#1	Pothole		91%	High	6.08%	Hot-mix / cold-mix asphalt fill with mechanical compaction
#2	Pothole		77%	Medium	2.85%	Edge trimming, tack coat and patch sealing
#3	Alligator Crack		58%	High	3.61%	Bituminous slurry seal over the cracked area
#4	Pothole		44%	Minor	0.88%	Localised surface levelling during the next maintenance round
#5	Alligator Crack		36%	Minor	1.12%	Crack sealing during routine maintenance

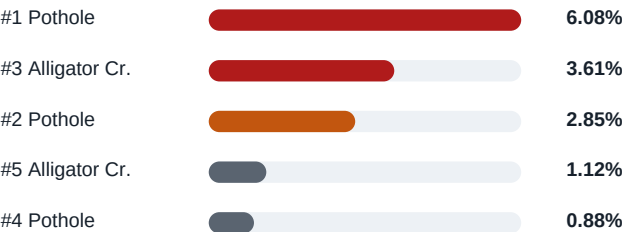
RISK ANALYTICS

How the index was arrived at

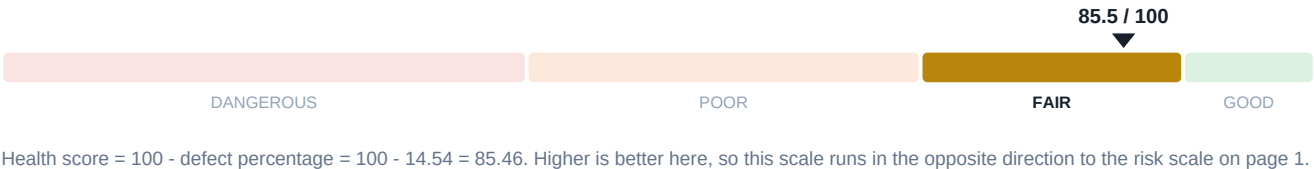
DEFECT COUNT BY SEVERITY BAND



FRAME SHARE PER DEFECT



ROAD HEALTH SCORE AGAINST THE PROJECT'S CONDITION BANDS



METHODOLOGY & LIMITATIONS

Detection. Trained object-detection model (best.pt), single-pass CPU inference at 640 px, confidence threshold 0.25, NMS IoU 0.45. Classes: pothole, alligator_crack.

Health score. Defect area was measured by the 'sum' method - summed bounding-box area 14.54%, overlap-corrected union area 14.54% - as a share of the 1600 x 1000 px frame. Road Health Score = 100 - defect percentage = 85.46. Higher is better here, so this scale runs in the opposite direction to the risk scale on page 1.

Risk index. $(0.50 \times \text{extent} + 0.30 \times \text{worst defect} + 0.20 \times \text{density}) \times \text{hazard} 0.89 \times \text{confidence} 0.61 = 32.43$, placing this location in the Moderate band (Priority-3, Remediate within 30 days). Severity of a single defect is banded by the share of the frame its box covers, and demoted one band when the model's confidence is below 0.40.

Limitations. Every threshold, weight and band used above - the Good/Fair/Poor/Dangerous score bands, the severity bands, the class hazard weights and the risk bands - is defined by this project team for the purpose of this tool. None of them is taken from IRC, MoRTH, GHMC or any other official standard, and the recommended remediation is a starting point for the engineering wing rather than an engineering directive. A photograph carries no depth and no scale, so 'severity' reflects how much of the frame a defect occupies, not how deep it is, and the same defect photographed from a different distance will score differently. This document is an automated visual estimate from a single photograph and does not replace physical inspection by a qualified engineer.

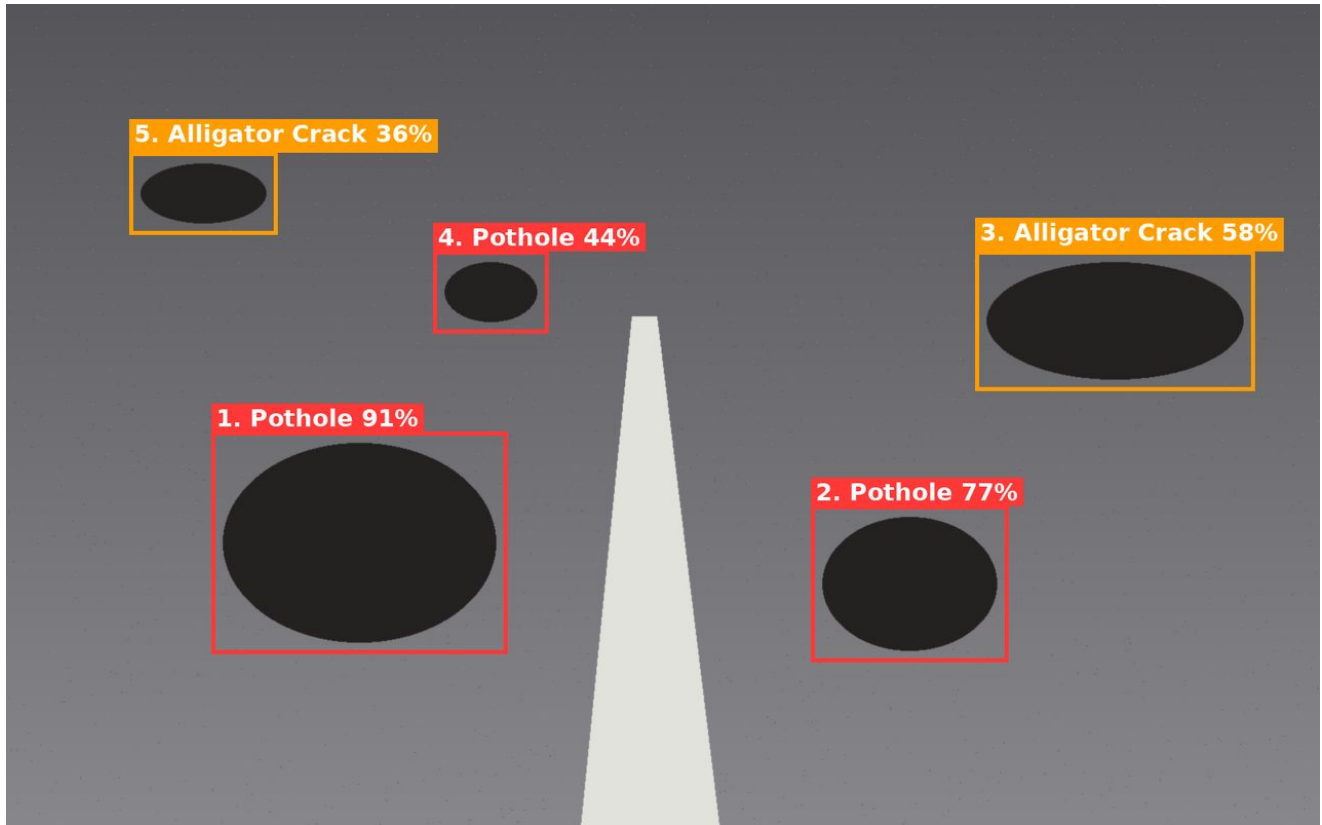
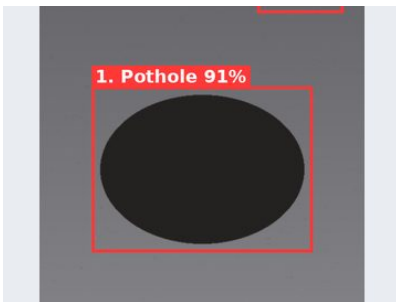


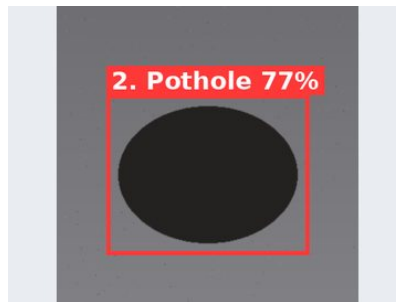
Figure 1. Detected defects with class and confidence. Each label number matches a tag in the inventory on page 2. 1280 x 800 px.

ISOLATED DEFECT CLOSE-UPS

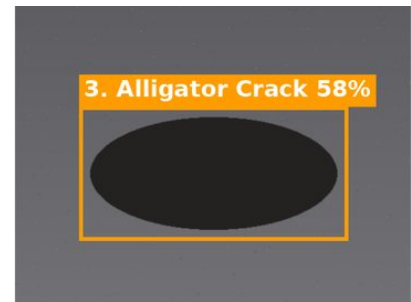
One crop per detection



Tag #1 · Pothole
High · 91% confidence



Tag #2 · Pothole
Medium · 77% confidence



Tag #3 · Alligator Crack
High · 58% confidence



Tag #4 · Pothole
Minor · 44% confidence



Tag #5 · Alligator Crack
Minor · 36% confidence